

Appointments:

- 2025 - 2027 **Postdoctoral Researcher**, *Radio Systems Research, Nokia Bell Labs, USA*
- 2024 **Graduate Teaching Assistant**, *Cornell University*, ECE 5670: Digital Communication System Design
- 2022 **Graduate Teaching Assistant**, *Cornell University*, ECE 2720: Data Science for Engineers
- 2019 **Undergraduate Teaching Assistant**, *Cornell University*, ECE 3100: Probability and Inference for Random Signals and Systems
- 2018 **Undergraduate Teaching Assistant**, *Cornell University*, ECE 2300: Digital Logic & Computer Organization

Education:

- 2015–2019 **Bachelor of Science**, *Cornell University*, Ithaca, NY, GPA: 4.077
Electrical and Computer Engineering
- 2019–2025 **PhD**, *Cornell University*, Ithaca, NY
Electrical and Computer Engineering

Awards and Honors:

- 2026 Winner of the 2026 Thomas M. Cover Dissertation Award
- 2025 Winner of the 2025 Cornell ECE Outstanding Ph.D. Thesis Research Award
- 2022 Jack Kiel Wolf ISIT Student Paper Award Finalist
- 2019 Graduated *Summa Cum Laude*
- 2018 Frank and Rosa Rhodes Scholarship recipient
- 2018 Nominated for the Outstanding Undergraduate Teaching Assistant Award
- [Cambridge International Qualifications \(Certificates\)](#)
- 2015 **GCE A Levels:**
 - Top in the World in A Level Physics
 - Top in the World in A Level Further Mathematics
 - Top in the World in A Level Chemistry
 - Top in Lahore in A Level Mathematics
 - First Place in Pakistan for Best Across Four Cambridge International A Levels
- 2013 **GCE O Levels:**
 - Top in the World in Additional Mathematics
 - First Place in Pakistan for Best Across Seven Cambridge International O Levels

Journal Publications:

- 2023 A. Mahmood and A. B. Wagner, "Lossy Compression With Universal Distortion," *IEEE Transactions on Information Theory*, vol. 69, no. 6, pp. 3552-3573, June 2023.

- 2023 A. Mahmood and A. B. Wagner, "Minimax Rate-Distortion," *IEEE Transactions on Information Theory*, vol. 69, no. 12, pp. 7712-7737, Dec. 2023.
- 2025 A. Mahmood and A. B. Wagner, "Channel Coding With Mean and Variance Cost Constraints," *IEEE Transactions on Information Theory*, vol. 71, no. 3, pp. 1504-1532, March 2025.
- 2025 A. Mahmood and A. B. Wagner, "Improved Channel Coding Performance Through Cost Variability," *IEEE Transactions on Communications*, vol. 73, no. 11, pp. 10145-10155, Nov. 2025.
- 2025 A. Mahmood and A. B. Wagner, "Channel Coding for Gaussian Channels With Mean and Variance Constraints," *IEEE Transactions on Information Theory*, vol. 71, no. 12, pp. 9285-9301, Dec. 2025.

Manuscripts under review:

- 2025 A. Mahmood and A. B. Wagner, "Channel Coding for Gaussian Channels with Multifaceted Power Constraints," *IEEE Transactions on Information Theory*, submitted Nov. 12, 2025.

Conference Publications:

- 2022 A. Mahmood and A. B. Wagner, "Lossy Compression with Universal Distortion," *2022 IEEE International Symposium on Information Theory (ISIT)*, Espoo, Finland, 2022, pp. 596-601.
- 2022 A. Mahmood and A. B. Wagner, "Minimax Rate-Distortion," *2022 IEEE International Symposium on Information Theory (ISIT)*, Espoo, Finland, 2022, pp. 602-607.
- 2023 A. Mahmood and A. B. Wagner, "Timid/Bold Coding for Channels with Cost Constraints," *2023 IEEE International Symposium on Information Theory (ISIT)*, Taipei, Taiwan, 2023, pp. 1442-1447.
- 2024 A. Mahmood and A. B. Wagner, "Channel Coding with Mean and Variance Cost Constraints," *2024 IEEE International Symposium on Information Theory (ISIT)*, Athens, Greece, 2024, pp. 510-515.
- 2025 A. Mahmood and A. B. Wagner, "Channel Coding for Gaussian Channels with Mean and Variance Constraints," *2025 IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, 2025, pp. 1-6.
- 2026 A. Mahmood and A. B. Wagner, "Cost Constraints Revisited," *2026 International Zurich Seminar on Information and Communication* (Invited).

Preprints and Technical Reports:

- 2026 A. Mahmood, "Weighted Unequal Error Protection over a Rayleigh Fading Channel," arXiv:2602.24225 [cs.IT], 2026.